WARG ROBOTICS



REAL WORLD, UNREAL EXPERIENCES

A camera you can build on.

Low-latency digital video, onboard recording and MAVLink telemetry on one encrypted link, with a companion computer on the way.

GLASS TO GLASS	RANGE	BAND	ENCRYPTION
60 ms	5 km	5.8 GHz	256-bit

IN THE FIELD

Three jobs it does today

What we have proven on each, and what is still ahead of us.



FPV and freestyle

Below about a hundred milliseconds the link stops being something a pilot compensates for.

- 60 ms through gates and tight lines
- Holds under hard banking and prop wash



Computer vision

For anything reactive, like obstacle avoidance or target tracking, round-trip latency decides whether it works.

- Stable link at extended standoff
- Onboard compute with the Warg SDK



Remote cockpit

Range and encryption matter most when the asset is far from the operator and the footage may be evidence.

- Survives sustained chassis vibration
- Low mounting height

BUILT AROUND YOUR CONSTRAINT

Imaging to your optical path

Lens and field of view chosen per airframe; the sensor swapped where the light demands it.

Chassis cut to your vehicle

Mount pattern, stack height, connector orientation and weight budget agreed before tooling.

Transport tuned to your site

Channel plan, bitrate and range matched to your link budget and regulatory ceiling.

An open feed

Standard output into your own pipeline. No proprietary decoder in the path.

CHOICE ONE

The camera

Three units on one link. They differ only in the picture and the storage.



Ghost JON

AVAILABLE NOW

Live and recording	720p · 60 fps
Sensor	8 MP Sony IMX219
Storage	8 GB ~50 MIN AT FULL BITRATE
Latency · range	60 ms · up to 5 km



Greywind ROBB

AVAILABLE NOW

Live and recording	1080p · 30 fps 720P AT 60 FPS
Sensor	8 MP Sony IMX219
Storage	16 GB ~1 H 45 AT FULL BITRATE
Latency · range	60 ms · up to 5 km



Raven BRAN

LAUNCHING SOON

Recording	4K · 30 fps PROVISIONAL
Storage	32 GB PROVISIONAL

SHARED BY EVERY UNIT

The link

A two-part 5.8 GHz digital link carrying video and two-way telemetry at the same time. Quality adapts to the signal, so the picture degrades gracefully at range instead of cutting out.



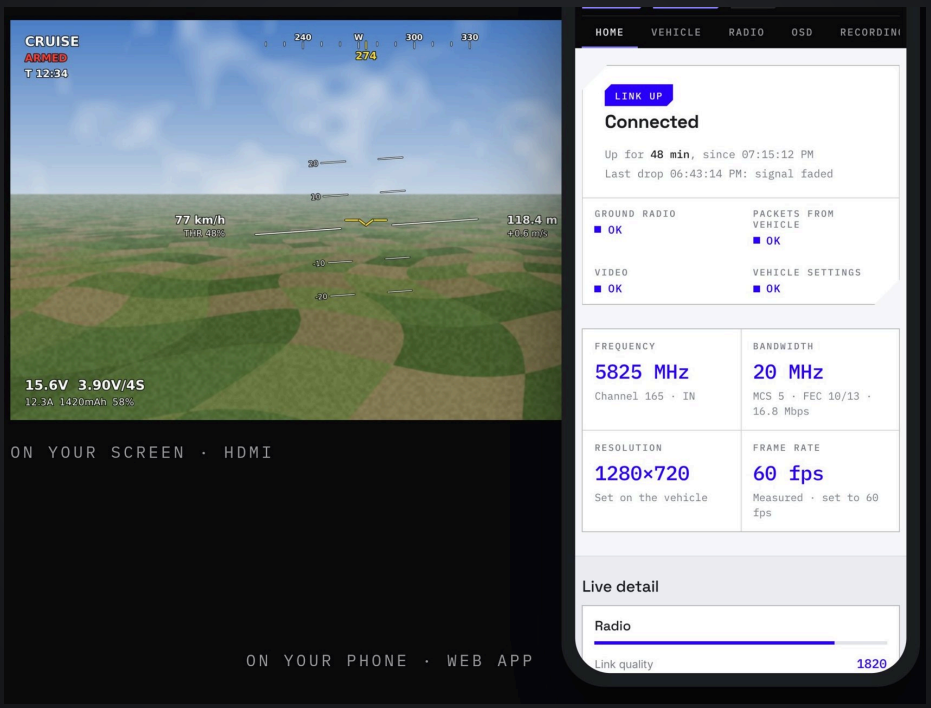
Frequency band	5725–5875 MHz
Channels	149, 153, 157, 161, 165, 169, 173
Channel width	20 MHz 40 MHZ ON 149, 157, 165, EXPERIMENTAL
Modulation	OFDM (802.11n-based), BPSK to 64-QAM
Antennas	2 per unit, 2T2R MIMO with diversity
Transmit power	1–25 dBm, adjustable DEFAULT 5 DBM
Video bitrate	1.5–19 Mbit/s, adaptive
On link loss	Drops to its most robust mode, recovers without a restart
Security	256-bit authenticated encryption VIDEO, TELEMETRY AND CONTROL
Pairing	One key per air and ground pair

The region setting enforces local channel rules; India is the default. The operator sets power to suit the antenna and stays within local EIRP limits (India: 36 dBm, 4 W).

GROUND STATION WEB APP · INCLUDED WITH EVERY UNIT

Your whole link, in any browser.

Join the ground unit's Wi-Fi, open one page, and see, set and record everything on both ends of the link. Nothing to install on a phone, tablet or laptop.



01 · WATCH

Link health, video and flight data, live.

02 · TUNE

Camera, radio, OSD and telemetry, from one page.

03 · KEEP

Record on the vehicle, download over Wi-Fi.

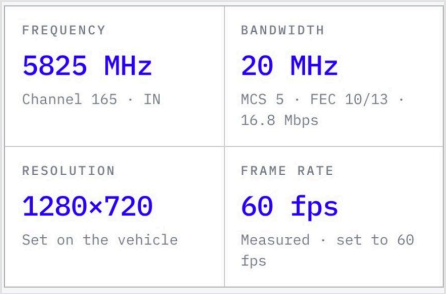
Connect in three steps: power both units, scan the first QR code on the screen to join the Wi-Fi, scan the second (or open 10.71.0.1).

WEB APP · HOME AND ALERTS

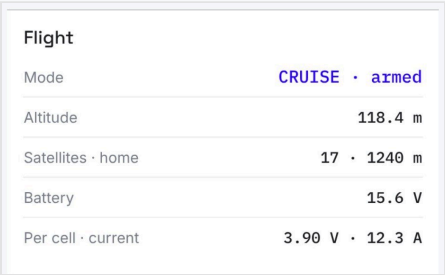
Know the link is healthy before you launch



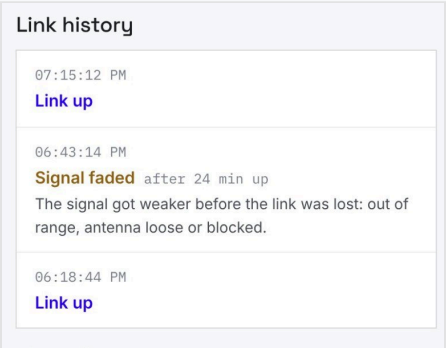
Link status. Four checks, ground radio to video. If one fails, it says why.



Now. Frequency, bandwidth, resolution and the frame rate actually received.



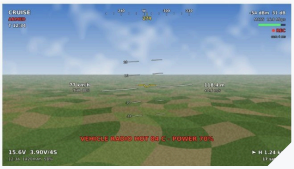
Flight data. Mode, altitude, GPS, distance home and battery from the flight controller.



Link history. Every drop with its likely cause: weak signal, power loss, wrong pairing.



Battery and link. Yellow warns, red is critical; flight controller messages too.



Radio heat. A hot radio lowers its own power to keep the link alive, and says so.

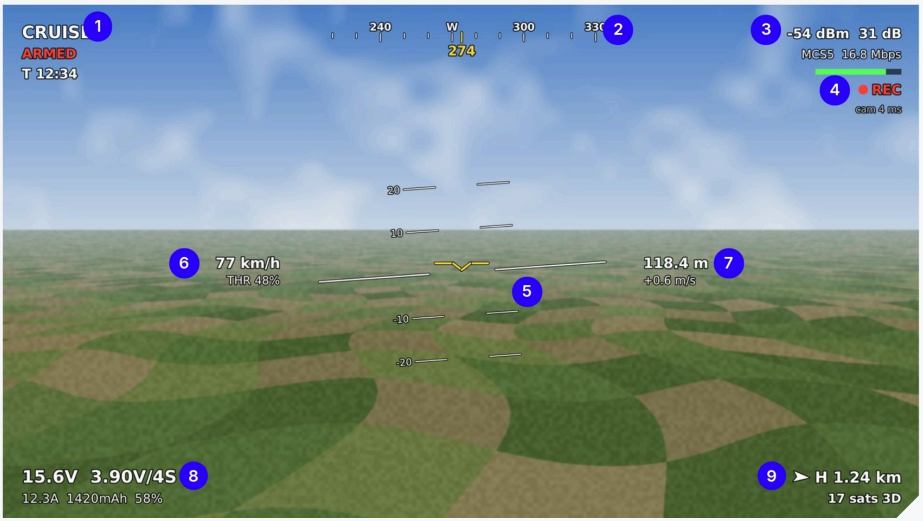


Status. Video lost, update progress and Wi-Fi details, on screen.

ON-SCREEN DISPLAY

Everything the operator needs, over the video

Flight data from the flight controller and link data from the radio, drawn on the live picture.



1 **Mode, armed, timer.** Flight mode and time in the air.

2 **Compass.** Heading tape and degrees.

3 **Link.** Signal, noise margin, bitrate and a quality bar.

4 **Recording, latency.** REC while recording; camera delay in ms.

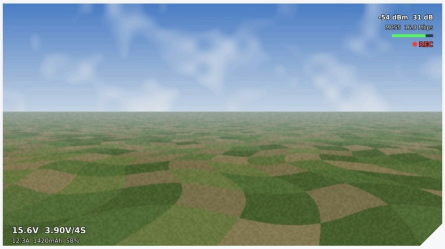
5 **Horizon.** Artificial horizon with pitch ladder.

6 **Speed, throttle.** Speed and throttle %.

7 **Altitude.** Height and climb rate.

8 **Battery.** Voltage, per cell, current, mAh used, %.

9 **Home, GPS.** Arrow and distance home; satellites and fix.



Show only what you need. Here: battery, link and REC.

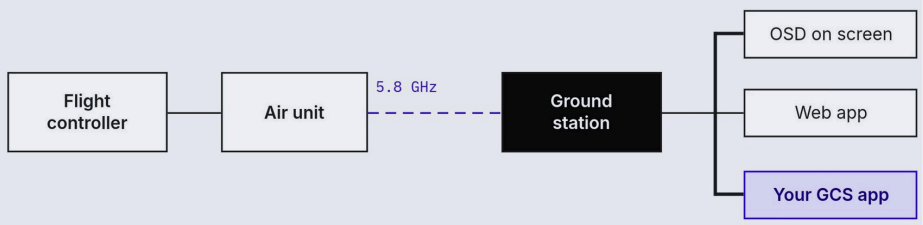
☑ mode	☑ battery
☑ link	☑ gps
☑ home	☑ altitude
☑ speed	☑ horizon
☑ compass	☑ timer
☑ messages	☑ rec
☑ throttle	☑ latency

OSD tab. Any of 14 items, metric or imperial, with your own battery warning levels.

INTEGRATION

Into your stack

Two-way MAVLink rides the same link as the video, so no extra telemetry radio is needed. The ground unit hands it to the OSD, the web app and your GCS.



On the hotspot. Point QGroundControl or Mission Planner at UDP port 14551. It connects by itself.

Watch or fly. Let the GCS control the vehicle (arm, modes, missions), or only watch.

On another network. Add the app's address under Send telemetry to.

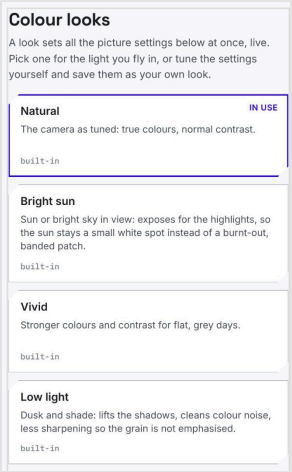
Safe by design. Only complete MAVLink messages reach the vehicle, and only from your own network.

	AIR UNIT	GROUND UNIT
Role	Camera, encoding, recording, transmitter	Receiver, display, setup hotspot
Camera input	MIPI CSI-2, camera included	—
Flight controller	UART, 3.3 V logic, to 921600 baud	—
Video out	—	HDMI, with OSD
Ground control	—	MAVLink over UDP, Wi-Fi or Ethernet
Local network	2.4 GHz Wi-Fi, off for flight if you want	2.4 GHz hotspot (WPA2), Ethernet
Antennas	2 × 5.8 GHz	2 × 5.8 GHz
Supply	5 V; radio on its own feed recommended	5 V
Thermal	Automatic protection, three stages	Automatic protection

WEB APP · VEHICLE, RADIO AND PAIRING

Configure both ends from one page

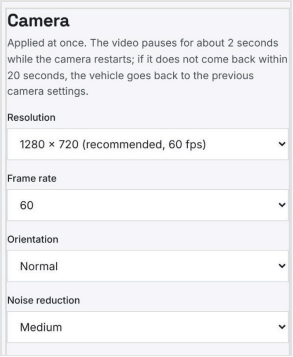
Changes show on the live video at once, and radio settings switch on the ground unit and the vehicle together.



Colour looks. Natural, Bright sun, Vivid, Low light, or your own.



Picture. Sharpness, colour, contrast, brightness, exposure, white balance.



Camera. Up to 1920 × 1080, frame rate, flip, noise reduction. A failed change reverts by itself.

Channel and bandwidth

Pick a 5.8 GHz channel and region. If the vehicle doesn't answer on the new channel within 35 s, both units go back by themselves.

Power and bitrate

Transmit power per unit, with what the radio really uses. Optionally cap the video bitrate.

Pairing

Press Pair near the vehicle; about a minute later the two are locked to each other. Only public keys are exchanged.

Flight controller port

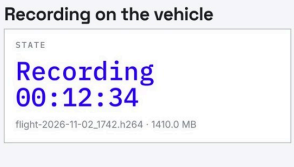
Pick the port and speed. The vehicle sets itself up. Focus assist helps set the lens in the field.

VIDEO MODES

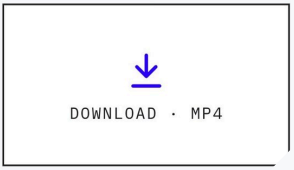
Selectable	1920 × 1080, 1640 × 1232, 1280 × 720, 1024 × 576, 960 × 720, 854 × 480, 640 × 480
Codec · bitrate	H.264 · 1.5–19 Mbit/s, adaptive

WEB APP · RECORDING AND SYSTEM

Record on the vehicle. Keep the fleet current.







1 · Start. Name the recording and press Start; timer and size count up.

2 · Fly. REC on the OSD confirms it.

3 · Download. MP4 files copy over Wi-Fi after landing.

Same quality as live

Saved as MP4 on the vehicle, at the live-stream resolution and bitrate.

Video keeps flowing

Downloads use Wi-Fi, never the video link, and resume if interrupted.

Never fills up

Recording stops cleanly before the storage runs out.

Every flight listed

Date, size and free space on the vehicle, in one list.

Safe updates

Both units update together. Only signed Warg releases install, never while armed, and a unit that doesn't come back healthy returns to its previous version.

Update channel

Automatic or one tap; stable or beta releases.

Page password

So only your operators can change settings.

Quick fixes

Restart the video or the link, or reboot either unit, with one button. Logs are a tap away.

CHOICE TWO, INDEPENDENT OF THE FIRST · LAUNCHING SOON

The compute

Every unit in the line will be available as a Pro: the same camera and the same link, with a Linux processor on board. Memory and storage are chosen separately from the camera.

MEMORY	STORAGE
2 / 4 / 8 GB	32 / 64 / 128 GB

WHAT YOU CAN RUN ON IT

Image processing

Detection and tracking against the full-rate frame, where it is captured.

Autonomous navigation

The aircraft acts on what it sees, so the link is no longer in the control loop.

Path finding

Routing computed from the live picture rather than a map made earlier.

Precision flight

Close-quarters work where ground-station latency is the limiting factor.

PLATFORM

Processor · OS	ARM 64 · Linux
Software	Warg SDK, container deployment
Protocol · connector	MAVLink · JST-SH
Input voltage	5.5–36 V

Get in touch

Tell us what you are building and we will take it from there. Bulk orders, OEM programmes, custom work and evaluation units all start the same way.



BUYING TODAY

Ghost	₹29,999 + GST
Greywind	₹35,999 + GST
Ghost + Greywind	₹49,999 + GST per pair
Pro, Raven	Launching soon; quoted on the configuration

Buying for a business? Add your GSTIN at checkout and the GST invoice carries it, for input tax credit. Up to 9 units per online order; for more, write to us.

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WARG ROBOTICS

DESIGNED AND MADE IN INDIA